MSc Technologies and Analytics in Precision Medicine

Module: Advanced Data Analytics

Assignment 3

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March 2026

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1 Introduction

Type 2 diabetes represents a significant global health challenge, affecting hundreds of millions of people worldwide. Identifying individuals at elevated risk is important for enabling early intervention and prevention strategies. The Pima Indians of Arizona have one of the highest documented rates of diabetes, making this population particularly valuable for investigating factors associated with the disease.

This study uses the 'Pima Indians Diabetes 2' dataset, which contains clinical measurements from 768 women aged 21 years or older, to explore predictors of diabetes diagnosis. The dataset includes eight continuous clinical variables and a binary outcome indicating whether an individual has diabetes.

The objective of this analysis is to determine which clinical and demographic factors are most predictive of diabetes. Logistic regression models are used to examine the predictive value of the original variables, while Lasso regression is applied to perform variable selection and evaluate the impact of model complexity. In addition, potential non-linear effects and interactions between predictors are considered to assess whether they improve predictive performance.

2 Methods

2.1 Data Preparation and Exploration

The 'Pima Indians Diabetes 2' dataset was loaded and observations with missing values were removed to obtain a complete-case dataset. Summary statistics were calculated for all variables. Correlations between predictor variables were assessed using a correlation matrix and visualised using a heatmap.

The cleaned dataset was randomly divided into training (70%) and test (30%) subsets using a fixed random seed to ensure reproducibility. The training data were used to fit all models, while the test data were reserved for evaluating predictive performance.

2.2 Full Logistic Regression Model

A logistic regression model was fitted on the training data using the original eight predictors. Model coefficients were examined and predicted probabilities were generated. Model performance was evaluated using the area under the receiver operating characteristic curve (AUC) on both the training and test datasets. A DeLong test was additionally performed to assess whether the difference between the training and test AUC values was statistically significant.

2.3 Expanded Predictor Model

To explore more complex relationships, an expanded predictor space was created by including squared (polynomial) terms for each predictor and all possible two-way interaction terms. A logistic regression model was then fitted using this expanded set of predictors. Model performance was assessed using AUC and a DeLong test was used to evaluate whether the difference between the two AUC values was statistically significant.

2.4 Lasso Regression

Lasso regression was applied to the expanded predictor matrix using the `cv.glmnet()` function with 10-fold cross-validation. The optimal regularisation parameters λ_{min} and λ_{1se} were identified. Coefficient paths and cross-validation curves were visualised to assess variable shrinkage and selection.

2.5 Model Comparison

Predictive performance was compared across four models: the simple logistic regression model, the expanded logistic regression model, and the Lasso models evaluated at λ_{min} and λ_{1se}. Test AUC values were calculated for each model and ROC curves were plotted to compare classification performance.

2.6 Interpretation of Selected Variables

Non-zero coefficients from the Lasso model at λ_{1se} were extracted to identify variables retained after regularisation. These coefficients were visualised using a bar chart to illustrate the predictors contributing to the final parsimonious model.

3 Results and Interpretation

3.1 Data Preparation and Exploration

Table 1 presents the summary statistics for the variables. The mean plasma glucose concentration was 121.69 mg/dL (SD = 30.54) and the mean body mass index (BMI) was 32.46 kg/m^2 (SD = 6.92), both of which are important metabolic indicators associated with diabetes risk (Zhao et al. 2024). Insulin levels showed substantial variability (mean = 155.55 µU/mL, SD = 118.78), suggesting considerable heterogeneity in metabolic function among participants. The diabetes pedigree function had a mean of 0.47 (SD = 0.33), reflecting variation in genetic susceptibility to diabetes (Goyal et al. 2023). Overall, 268 of the 768 participants (34.9%) were classified as diabetic, while 65.1% were non-diabetic.

Variable	n	Mean	SD
Pregnant	392	3.30	3.21
Glucose	392	122.63	30.86
Pressure	392	70.66	12.50
Triceps	392	29.15	10.52
Insulin	392	156.06	118.84
Mass (BMI)	392	33.09	7.03
Pedigree	392	0.52	0.35
Age	392	30.86	10.20
Diabetes	392	1.33	0.47

Table 1: Summary statistics for the cleaned Pima Indians Diabetes dataset after removing observations with missing values.

Figure 1 shows the correlation heatmap of the predictor variables. Most correlations between predictors were weak to moderate, suggesting limited multicollinearity. The strongest positive correlation was observed between triceps skinfold thickness and BMI (mass), which is expected as both measure body adiposity (Mukkamala et al. 2021). A moderate positive correlation was also observed between glucose and insulin, reflecting their biological relationship in glucose metabolism. Additionally, age and number of pregnancies showed a noticeable positive correlation, likely reflecting the accumulation of pregnancies with increasing age. Overall, no extremely high correlations were observed, indicating that the predictors provide largely distinct information for modelling diabetes risk.

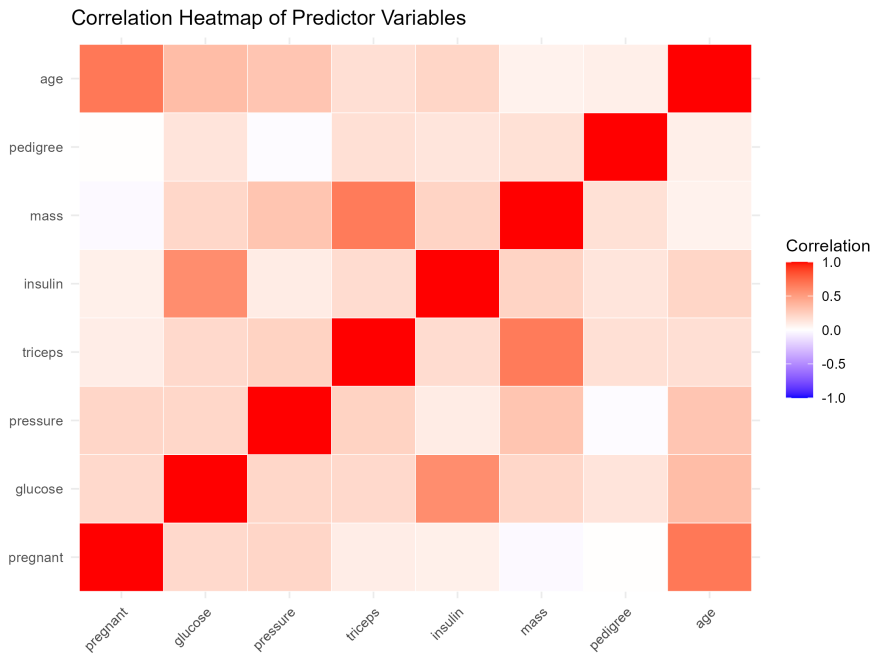

Figure 1: Figure 1: Correlation heatmap of predictor variables

3.2 Full Logistic Regression Model

Table 2 presents the AUC values for the logistic regression model on the training and test datasets. The model achieved an AUC of 0.8768 on the training set and 0.8034 on the test set, indicating good discriminatory ability with slightly lower performance on unseen data. Overfitting occurs when a model captures noise in the training data rather than the underlying signal, resulting in substantially poorer performance on the test dataset. Although the test AUC is lower than the training AUC, the difference is modest. A DeLong test comparing the two ROC curves yielded a p-value of 0.1508, indicating that the difference in AUC is not statistically significant. Therefore, there is little evidence that the model is severely overfitted.

Dataset	AUC
Training set	0.8768
Test set	0.8034

Table 2: Area Under the Curve (AUC) for the logistic regression model on the training and test datasets.

3.3 Expanded Predictor Space

Figure 2 illustrates the R code used to create the expanded predictor matrix that was created to allow the model to capture more complex relationships between variables. In addition to the eight original predictors, squared polynomial terms were included to account for potential non-linear effects, and all two-way interaction terms between predictors were added. This resulted in an expanded predictor space consisting of 44 variables.

```r
###TASK 3:

#creating squared terms for training data
train_expanded <- train_data %>%
  mutate(
    pregnant2 = pregnant^2,
    glucose2 = glucose^2,
    pressure2 = pressure^2,
    triceps2 = triceps^2,
    insulin2 = insulin^2,
    mass2 = mass^2,
    pedigree2 = pedigree^2,
    age2 = age^2
  )

#creating two-way interaction terms for training data
X_interactions <- model.matrix(
  ~ (pregnant + glucose + pressure + triceps + insulin + mass + pedigree + age)^2,
  data = train_expanded
)[, -1]

#combining to create expanded predictor space for training data
X_train <- cbind(
  X_interactions,
  train_expanded[, c("pregnant2","glucose2","pressure2","triceps2",
                     "insulin2","mass2","pedigree2","age2")]
)
dim(X_train)
```

Figure 2: R code used to create expanded matrix

Table 3 shows the AUC values for the logistic regression model fitted on the expanded predictor space. The training AUC (0.9301) is substantially higher than the test AUC (0.7312), suggesting

that the model may be overfitting the training data. A DeLong test comparing the two ROC curves indicated that this difference was statistically significant ($p < 0.001$). When the number of predictors becomes large relative to the sample size, standard logistic regression models can become unstable.

In this analysis, the inclusion of squared terms and all two-way interactions increased the number of predictors to 44, substantially increasing model complexity. Logistic regression assumes linear and additive relationships, and introducing numerous interaction and polynomial terms can lead to highly flexible models that capture noise rather than the underlying signal. A commonly cited rule of thumb recommends at least 10–20 observations per predictor to obtain reliable estimates. Although the dataset contains 392 observations, fitting 44 predictors approaches this limit and increases the risk of overfitting, resulting in unstable coefficients and reduced generalisability to new data.

Dataset	AUC
Training set	0.9301
Test set	0.7312

Table 3: Area Under the Curve (AUC) for the logistic regression model.

3.4 Lasso Regression

Figure 3 shows the cross-validation curve for the lasso regression model, where the mean cross-validated binomial deviance is plotted against $\log(\lambda)$. The minimum cross-validated error occurs at $\lambda_{\text{min}} = [0.006]$, while the $\lambda_{\text{1se}} = [0.113]$ corresponds to the most parsimonious model within one standard error of the minimum.

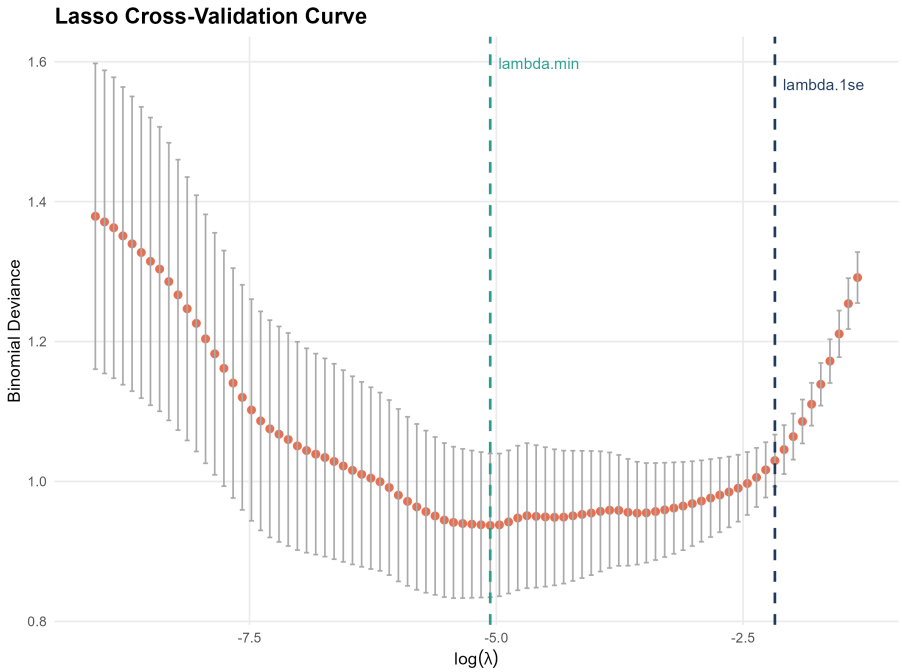

Figure 3: Cross-validation curve for the Lasso logistic regression model.

Figure 4 shows the coefficient paths for the lasso regression model as the regularisation parameter λ increases. As λ increases (moving right along the x-axis), the coefficients shrink towards zero, with many predictors being eliminated from the model. Only a small number of predictors remain non-zero at larger λ values, indicating the variable selection effect of the lasso penalty.

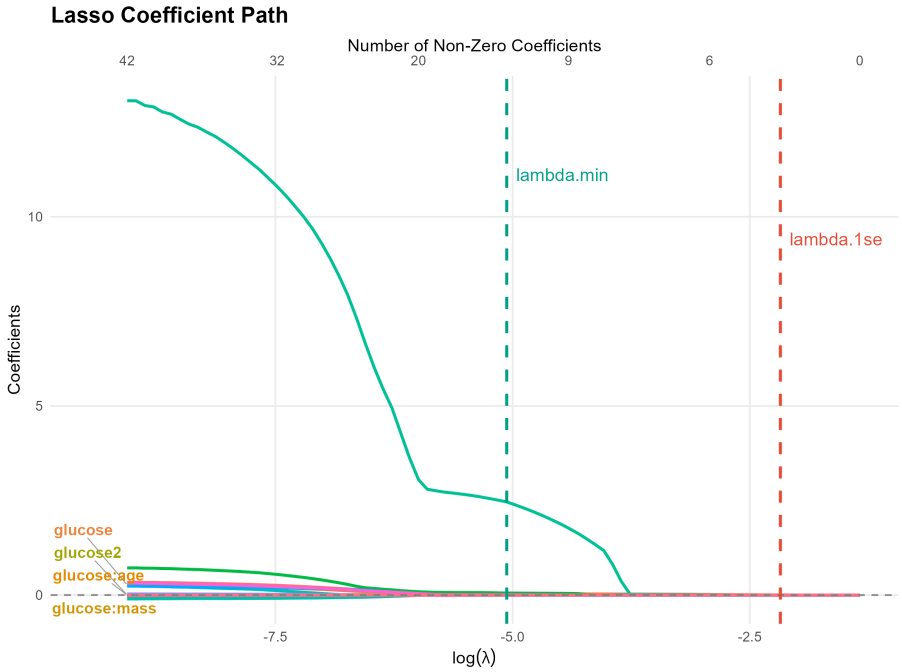

Figure 4: Lasso Coefficient Paths

Table 4 and *Table 5* present the non-zero coefficients selected by the lasso model at $\lambda_{\min}$ and $\lambda_{1\text{se}}$, respectively. At $\lambda_{\min}$, the model retained 11 predictors, including the main effects of glucose, triceps skinfold thickness, BMI (mass), and the diabetes pedigree function, along with several interaction and polynomial terms.

In contrast, the more parsimonious model obtained at $\lambda_{1\text{se}}$ retained only four predictors: glucose, the interaction terms glucose×mass and glucose×age, and the squared term glucose^2. This reduction in the number of predictors reflects the variable selection property of the lasso penalty, which penalises the absolute values of the regression coefficients and shrinks many of them exactly to zero as the regularisation parameter increases

Predictor	Coefficient
glucose	0.03297
triceps	0.00755
mass	0.05323
pedigree	2.46171
pregnant:insulin	6.76e-05
pressure:age	1.36e-05
insulin:pedigree	-0.00612
insulin:age	9.67e-05
mass:age	4.15e-04
glucose^2	4.34e-05
insulin^2	-4.65e-10

Table 4: Non-zero coefficients selected by the Lasso model at $\lambda_{\min}$ (11 predictors).

Predictor	Coefficient
glucose	0.003522
glucose:mass	2.55e-04
glucose:age	6.28e-05
glucose^2	2.14e-05

Table 5: Non-zero coefficients selected by the Lasso model at $\lambda_{1\text{se}}$ (4 predictors).

3.5 Model Comparison

Table 6 summarises the test AUC values for the four models considered: the simple logistic regression model with eight predictors, the expanded logistic regression model with 44 predictors, and the Lasso models evaluated at $\lambda_{\min}$ and $\lambda_{1\text{se}}$. The table also reports the number of predictors retained in each model.

Model	Number of Predictors	Test AUC
Simple	8	0.8034
Full	44	0.7312
Lasso (λ_{min})	11	0.8124
Lasso (λ_{1se})	4	0.7835

Table 6: Comparison of model performance on the test dataset.

Figure 5 shows the ROC curves for the four models evaluated on the test dataset: the simple logistic model, the expanded logistic model, and the Lasso models at λ_{min} and λ_{1se}. The curves illustrate the trade-off between sensitivity and specificity across different classification thresholds, with the legend reporting the corresponding AUC values for each model.

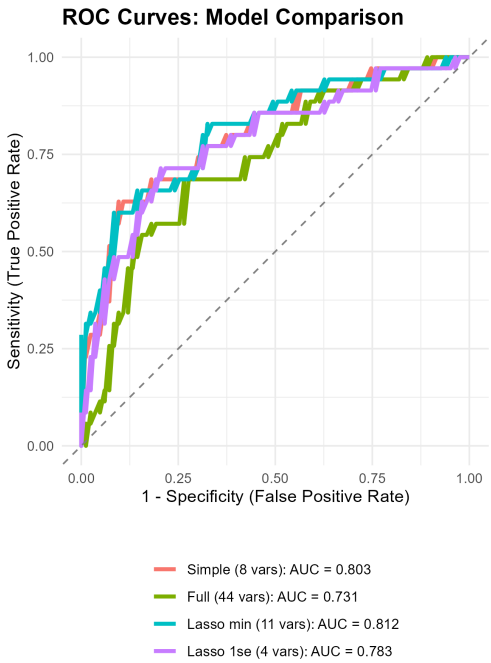

Figure 5: ROC Curve Comparison of Model Performance

3.5.1 Interpretation

Based on the test AUC values, the lasso model evaluated at λ_{min} performed best, achieving an AUC of 0.812. In contrast, the expanded logistic regression model with 44 predictors performed worst, with an AUC of 0.731. This likely reflects overfitting, as the large number of predictors increases model complexity and variance, reducing the model's ability to generalise to unseen data. The bias–variance trade-off describes the balance between model simplicity and flexibility: overly simple models have high bias and may underfit the data, while overly complex models have high variance and may overfit.

Lasso regression addresses this issue by applying a regularisation penalty that shrinks coefficient

estimates and can set some coefficients exactly to zero, reducing model complexity. The model at λ_{min} retains more predictors and provides the best predictive performance, while the model at $\lambda_{1\text{se}}$ is more parsimonious but slightly less accurate. Since the objective of this analysis is predictive performance, the Lasso model at λ_{min} represents the best-performing model.

3.6 Clinical Interpretation

Figure 6 displays the non-zero coefficients selected by the Lasso model at $\lambda_{1\text{se}}$. Four predictors were retained in the model: glucose, the interaction terms glucose×mass and glucose×age, and the squared term glucose^2.

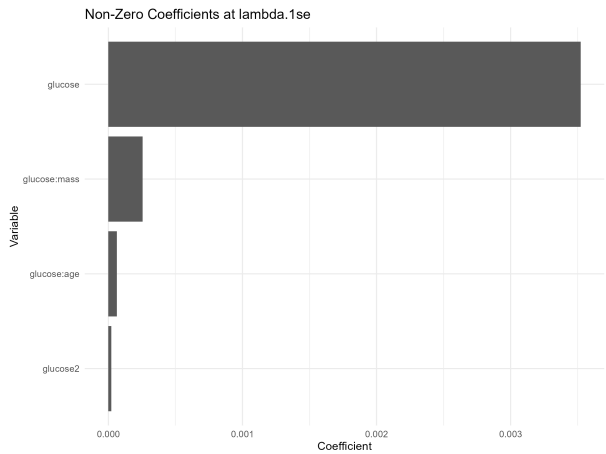

Figure 6: Bar chart showing the non-zero coefficients selected by the Lasso model at $\lambda_{1\text{se}}$.

The positive coefficient for glucose indicates that higher plasma glucose levels are strongly associated with an increased risk of diabetes. Elevated blood glucose reflects impaired glucose regulation and reduced insulin sensitivity, both of which are central features in the development of type 2 diabetes (*Pre-diabetes* 2026).

The interaction term glucose×mass suggests that the effect of glucose on diabetes risk may be stronger in individuals with higher body mass. Increased body mass index (BMI), particularly excess adipose tissue, contributes to insulin resistance and impaired glucose metabolism, thereby amplifying the effect of elevated glucose levels (Haines et al. 2022).

The interaction term glucose×age indicates that the relationship between glucose levels and diabetes risk may vary with age. As individuals age, physiological changes such as increased insulin resistance, reduced physical activity, and increased adiposity can worsen glucose tolerance, leading to a higher probability of diabetes (Chia, Egan, and Ferrucci 2018).

Finally, the inclusion of the polynomial term glucose^2 suggests a non-linear relationship between glucose and diabetes risk. This implies that risk may increase more rapidly at higher glucose concentrations, reflecting the progressive deterioration of glucose regulation as hyperglycaemia worsens (Cao et al. 2024).

3.6.1 Clinical Implications of Model

The results highlight plasma glucose as the dominant predictor of diabetes risk. In addition to glucose alone, the model indicates that its effect may be modified by body mass index (BMI) and age, suggesting that diabetes risk should be evaluated within a broader metabolic and demographic context rather than considering glucose levels in isolation.

These findings suggest that clinicians should prioritise monitoring plasma glucose alongside indicators such as BMI and patient age when assessing diabetes risk. Higher BMI is strongly linked to insulin resistance due to increased adipose tissue, while ageing is associated with physiological changes that reduce glucose tolerance (Wilcox 2005). Considering these factors together may support more personalised risk assessment and earlier identification of individuals at high risk.

No particular unexpected predictors were identified, as the selected variables align well with established mechanisms of type 2 diabetes development, particularly the relationship between hyperglycaemia, adiposity, and insulin resistance.

4 Conclusion

This study aimed to identify which clinical and demographic factors, including potential non-linear effects and interactions, are most predictive of diabetes in Pima Indian women. Across the models evaluated, plasma glucose emerged as the strongest predictor of diabetes risk. The lasso regression model further highlighted that the relationship between glucose and diabetes risk may be modified by body mass index (BMI) and age, and that the association between glucose and diabetes risk may be non-linear.

In terms of predictive performance, the lasso model evaluated at λ_{min} achieved the highest test AUC, indicating the best overall discrimination between diabetic and non-diabetic individuals. This result demonstrates the advantage of regularised models when working with a large predictor space, as Lasso reduces model complexity and mitigates overfitting by shrinking many coefficients to zero.

Several limitations should be considered. First, the complete-case analysis reduced the sample size from the original dataset to 392 observations, which may limit statistical power. In addition, the mean age of participants in the dataset was relatively young (approximately 31 years), whereas the onset of type 2 diabetes typically occurs later in life (*Diabetes Type 2* 2026), which may limit the generalisability of the findings to older populations. Future studies using larger and more age-diverse cohorts may provide a more comprehensive understanding of diabetes risk factors.

References

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